

TERRY TAEWOONG UM

Head of AI Innovation Division (VP), Cosmax · Physical AI for Manufacturing

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SUMMARY

I build Physical AI systems that work in the real world. I started as an AI & robotics researcher, publishing 20+ papers cited over 2,000 times, then founded ART Lab to bring these technologies into everyday industries. The company was acquired by Cosmax, the world's leading cosmetics manufacturer, where I now lead AI & Robotics — applying Physical AI to manufacturing systems in which data is messy, environments are dynamic, and solutions must actually work.

RESEARCH INTERESTS

Physical AI reshaping industrial environments; the AI transformation of manufacturing; and tech startups that bring technology into real-world industries — grounded in robotics, deep learning, and real-world AI systems.

EXPERIENCE

Cosmax — Vice President, Head of AI Innovation Division

Nov 2024 - Present

Joined through the acquisition of ART Lab. Leading Cosmax's transformation around AI & Robotics Technologies (ARTs), with a focus on Physical AI research for manufacturing.

AI Future Forum, FEELSCI — Co-chair

Honorary

Present

FEELSCI (과실연), founded in 2005, is the "Citizens' Coalition for a Sound Science and Technology Society"; its AI Future Forum gathers Korea's leading AI experts from academia and industry to broaden AI adoption and advance the technology ecosystem.

ART Lab — Founder & CEO

May 2019 - Nov 2024

Built a data-driven personalization system for the beauty industry. Started with seed investment from Naver and Cosmax; acquired by Cosmax in 2024.

- **SkinLog** — skincare diary app generating structured skin-cosmetic data
- **Skinyzer** — vision-based skin analysis & product recommendation module
- **SkinChat** — AI skin-consultation assistant combining vision and dermatology
- Established an end-to-end data flow between skin, AI, and cosmetics

University of Waterloo — PhD Researcher

Sep 2014 - Dec 2018

Research on human/robot motion understanding using deep learning, with applications in healthcare and wearable systems.

- Developed deep learning models for real-world sensor data (e.g., Parkinson's disease monitoring)
- Collected and worked with noisy, real-world datasets from hospitals and industry partners

- Published 20+ papers cited over 2,000 times

KIST (Korea Institute of Science and Technology) — Research Scientist

Nov 2012 – Aug 2014

- Researched transfer learning methods for robotic systems
- Developed assistive robotic systems for human motion support

LIG Nex1 — Research Engineer

Jan 2010 – Nov 2012

- Developed autonomous navigation systems for unmanned ground vehicles
- Designed robotic and exoskeleton systems for mobility and assistance

EDUCATION

University of Waterloo — Ph.D. (incomplete), Department of Electrical and Computer Engineering

2014 – 2022

Research: Deep Learning for Human/Robot Motions · Advisor: Dana Kulić

* Completed PhD coursework and the Comprehensive Examination; voluntary withdrawal in 2022, PhD degree not conferred.

Seoul National University — M.S. in Mechanical & Aerospace Engineering (Robotics)

2008 – 2010

Thesis: "Tangent Space RRT with Lazy Projection for Constrained Motion Planning" · Advisor: Frank C. Park

Seoul National University — B.S. in Mechanical & Aerospace Engineering

2002 – 2008

SELECTED PUBLICATIONS

20+ papers · 2,066 citations · h-index 11 · i10-index 11 – Google Scholar, last updated June 2026

1. **TT Um**, FMJ Pfister, D Pichler, S Endo, M Lang, S Hirche, U Fietzek, et al. "Data augmentation of wearable sensor data for Parkinson's disease monitoring using convolutional neural networks." *Proc. 19th ACM Int. Conf. on Multimodal Interaction (ICMI)*, 2017. (1,186 cites)
2. JY Choi, TK Yoo, JG Seo, J Kwak, **TT Um**, TH Rim. "Multi-categorical deep learning neural network to classify retinal images: A pilot study employing small database." *PLoS ONE*, 2017. (334 cites)
3. **TT Um**, V Babakeshizadeh, D Kulić. "Exercise motion classification from large-scale wearable sensor data using convolutional neural networks." *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2017. (141 cites)
4. B Kim, **TT Um**, C Suh, FC Park. "Tangent bundle RRT: A randomized algorithm for constrained motion planning." *Robotica*, 2016. (126 cites)
5. FMJ Pfister, **TT Um**, DC Pichler, J Goschenhofer, K Abedinpour, M Lang, et al. "High-resolution motor state detection in Parkinson's disease using convolutional neural networks." *Scientific Reports*, 2020. (89 cites)
6. C Suh, **TT Um**, B Kim, H Noh, M Kim, FC Park. "Tangent space RRT: A randomized planning algorithm on constraint manifolds." *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2011. (52 cites)
7. **TT Um**, B Kim, C Suh, FC Park. "Tangent space RRT with lazy projection: An efficient planning algorithm for constrained motions." *Advances in Robot Kinematics (ARK)*, 2010. (37 cites)

8. **TT Um**, MS Park, JM Park. "Independent joint learning: A novel task-to-task transfer learning scheme for robot models." *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2014. (28 cites)

OTHER PUBLICATIONS

9. **TT Um**, FMJ Pfister, DC Pichler, S Endo, M Lang, S Hirche, U Fietzek, et al. "Parkinson's disease assessment from a wrist-worn wearable sensor in free-living conditions: Deep ensemble learning and visualization." *arXiv:1808.02870*, 2018.
10. M Lang, FMJ Pfister, J Fröhner, K Abedinpour, D Pichler, U Fietzek, et al. (incl. **TT Um**). "A multi-layer Gaussian process for motor symptom estimation in people with Parkinson's disease." *IEEE Trans. Biomedical Engineering*, 2019.
11. MB Hong, SJ Kim, **T Um**, K Kim. "KULEX: An ADL power-assistance demonstration." *Int. Conf. on Ubiquitous Robots and Ambient Intelligence (URAI)*, 2013.
12. SI Cho, D Kim, H Lee, **TTW Um**, H Kim. "Explore highly relevant questions in the Baumann skin type questionnaire through the digital skin analyzer: A retrospective single-center study in South Korea." *Journal of Cosmetic Dermatology*, 2023.
13. **TT Um**, D Kulić. "An unsupervised approach to detecting and isolating athletic movements." *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2016.
14. S Endo, FJ Pfister, J Fröhner, U Fietzek, D Pichler, K Abedinpour, **TT Um**, et al. "Dynamics-based estimation of Parkinson's disease severity using Gaussian Processes." *IFAC Conf. on Cyber-Physical & Human Systems*, 2018.
15. F Pfister, D Kulić, **T Um**, D Pichler, A Ahmadi, M Lang, G König, F Achilles, et al. "Deep learning in objective classification of spontaneous movement of patients with Parkinson's disease using large-scale free-living sensor data." 2017.
16. JS Lee, B Han, H Lee, D Moon, D Kim, **TTW Um**, SI Cho, JI Na. "Robustness of automated acne lesion detection across various imaging conditions: A comparative analysis with dermatologists." *Journal of the European Academy of Dermatology & Venereology (JEADV)*, 2025.
17. F Pfister, J Goschenhofer, **T Um**, D Pichler, K Abedinpour, D Kulić, et al. "Quantification of the PD motor syndrome using an inclusive score (PD9) derived from sensor data with deep learning." *Movement Disorders*, 2018.
18. D Pichler, M Lang, FMJ Pfister, **T Um**, GCS König, S Endo, K Abedinpour, et al. "Untersuchung und Beurteilung der motorischen Symptome der Parkinson-Erkrankung mittels eines am Handgelenk getragenen Sensors und eines Bewegungserfassungssystems." *DGN-Kongress*, 2017.
19. D Pichler, M Lang, D Kulić, F Pfister, G Koenig, **T Um**, A Ahmadi, S Endo, et al. "Acquisition, validation and preprocessing of wrist-worn sensor data in patients with Parkinson's disease and healthy controls." 2017.
20. 엄태웅, 유재관, 백주현. "보병의 근력지원을 위한 하지 착용로봇 시스템 개발." *한국정밀공학회 학술발표대회 논문집*, 2012.

BOOKS

- "What I Wish I Had Known in Graduate School" (대학원생 때 알았더라면 좋았을 것들) — co-author; T. Um, Y. Choi, C. Kwon, 2019.
20th printing · ranked #1–3 in KAIST library loans (2020–2025)
- "Kakao AI Report" (카카오 AI 리포트) — contributing author, 2018.

- "AI and Deep Learning" (인공지능과 딥러닝) — technical reviewer, 2015.

SELECTED MEDIA

PRESS

[Terry Um, Cosmax — "AI and Robotics Will Take Global Beauty Higher"](#) — ETNews *(in Korean)* 2025

[Terry's interview, Cosmax](#) — Mael Business 2025

[Cosmax Acquires AI Startup ART Lab](#) — News1 *(in Korean)* 2024

[Terry Um, ART Lab — "AI Will Help You Find Better Skin Care"](#) — IT Donga *(in Korean)* 2024

TALKS & VIDEO

[ART Lab — Doing Skincare with Data](#) — YouTube · MSIT *(in Korean)* 2021

[Why Does the Skincare Market Need AI?](#) — YouTube · Terry TaeWoong Um *(in Korean)* 2021

[KAIST AI+Education: Education in the Age of AI](#) — YouTube · KAIST School of Computing *(in Korean)* 2019

LANGUAGES

Korean (Native) · English (Professional Working)